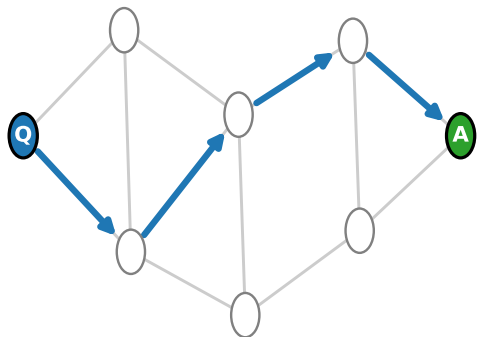


classical CoNet: transition probabilities,
one trajectory per attempt

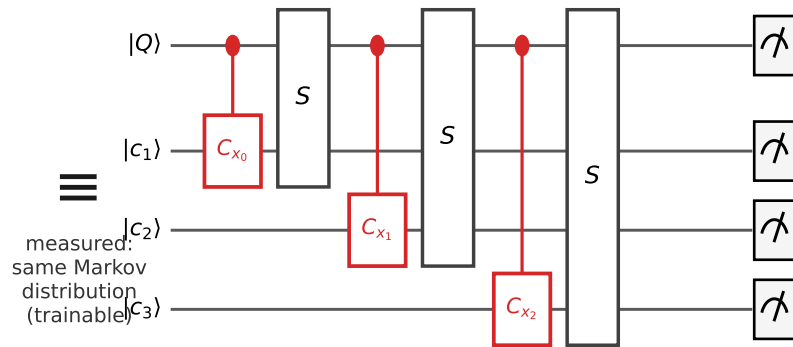


policy $\pi(\cdot x)$		
.62	.25	.13
.08	.81	.11
.33	.34	.33

sample one action at each step;
one attempt follows one trajectory

QuCoNet reasoning circuit:

$$\mathcal{A} = \prod_s C_x$$



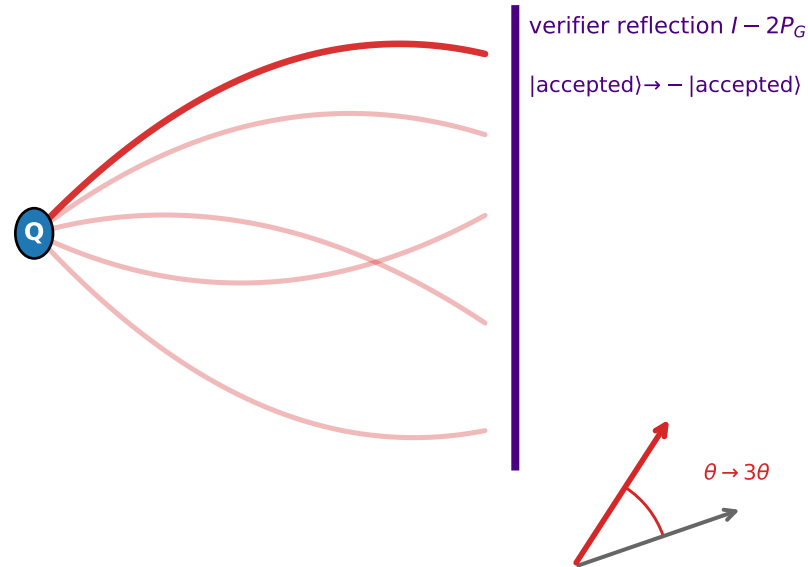
... M steps, one fresh path-record register each

the records mark which edge was taken; measured,
the walk is a classical Markov chain, its paths orthogonal

coherent verifier and state-preparation reflections:
Grover rounds amplify the accepted component



run
coherently:
verifier reflects



amplitude amplification: constant success in $O(1/\sqrt{p})$ coherent
queries where classical sampling needs $O(1/p)$ with terminal-outcome-only sampling